QUANTUM MEMORY FROM RARE THERMAL BOUNDARIES

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Abstract. A logical observable can change slowly because its commutator with the Hamiltonian is supported on a thermally rare subspace. Turning this observation into a memory theorem requires control of the state after an unknown input has been written. We give two complete criteria for that step under the exact joint system–bath dynamics. If all three logical Pauli commutators are supported on a common projection of Gibbs probability p, with summed norm bound $\mathfrak{a}$, ideal subsystem replacement gives deterministic half-diamond error at most $\mathfrak{a}p|t|F(\beta\mathfrak{a}/4)$, where $F(K) = \inf_{\lambda>0}(K + e^{\lambda} - 1)/\lambda$. A Pauli-randomized protocol instead has error at most $\mathfrak{a}p(2|t| + \beta)/4$. The proofs use rare-event relative entropy and a Bell-projection identity, respectively. The latter identity also gives a dimension-independent correlation criterion for arbitrary correlated resources and arbitrary joint quantum channels, including an algebraic infinite-bath formulation. We identify precise sufficient conditions for imperfect writing and prove that global domination by a thermal encoder can fail exponentially even for harmless independent gauge faults. All storage statements assume a specified accessible logical algebra and a fixed joint resource; preparation cost and a generic noisy-writing threshold are not supplied. The results are abstract criteria, rather than a new three-dimensional memory construction.

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Date: 6 November 2026.

2020 *Mathematics Subject Classification.* 81P45, 82B10, 81P48, 46L55.

Key words and phrases. quantum memory, Gibbs state, exact Hamiltonian dynamics, relative entropy, quantum channels, noiseless subsystems.

1. Introduction

A passive quantum memory must preserve an unknown state, including its entanglement with a reference, without active correction. Thermal equilibrium can help when logical errors must pass through configurations of small Gibbs weight. But an unknown input is not an equilibrium state. A correlation estimate becomes a memory theorem only after writing, holding and reading operations have been specified.

This paper studies that passage. Its basic hypothesis is an exact operator identity. The accessible physical system contains a full logical qubit algebra, and each logical Pauli commutator with the *actual joint Hamiltonian* is supported on a common projection P. We call P a thermal boundary: outside its range the instantaneous logical derivative vanishes. This name does not assert that P is a geometric surface, an energy eigenspace or an invariant subspace. In particular, P need not commute with the Hamiltonian or its Gibbs state.

The main theorem turns the equilibrium probability

$$p = \operatorname{Tr}(\omega P), \qquad \omega = \frac{e^{-\beta H}}{\operatorname{Tr} e^{-\beta H}},$$

into a bound on the entire memory channel. An ideal logical swap retains the joint marginal of the other physical variables and bath. Its relative-entropy cost, on a maximally mixed input, is $O(p)$. A binary entropy estimate keeps the boundary probability $O(p)$ at all times, and integration gives logical error $O(p|t|)$. A second proof uses thermal correlations and a classical Pauli frame. Both concern exact joint dynamics.

1.1. Relation to established approaches.

The correlation-and-fidelity framework is established. Alicki and Fannes [1], Sections 2–4, treat encoded matrix algebras, correlated invariant resources and CP dynamics, including logical replacement with a fixed syndrome marginal. The replacement used here belongs to that framework.

Theorem 1 of Chesi, Loss, Bravyi and Terhal [5] already bounds memory error by t times a bad-syndrome Gibbs probability for Gibbs-stationary Markovian dynamics of stabilizer and subsystem codes. Inverse-rarity storage is therefore a known mechanism. Our hypotheses instead concern total-Hamiltonian commutators and the interacting system–bath Gibbs state.

Balasubramanian, Davydova and Lin [3], Theorems 3.16 and 5.13 and Corollary 5.14, give a three-dimensional local stabilizer construction with long thermal storage under their stationary dynamics. Bergamaschi, Gheissari and Liu [4], Theorem 1.7 and formal Theorem 7.8, address thermalization within a logical sector under additional code and sampler hypotheses.

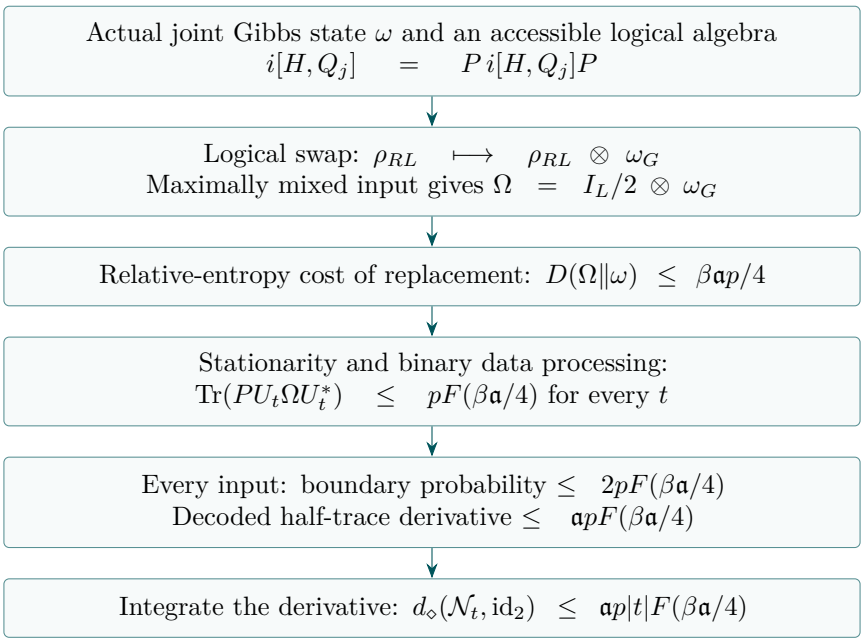
Rakovszky, Placke, Breuckmann and Khemani [14], Theorem 2 and Supplement II, control mixing through quantum bottlenecks, including off-diagonal contributions. Slow mixing alone does not preserve a coherent qubit. Ma, Khemani and Sang [9], Theorems 1–3, relate phase transport and self-correction under stronger clustering and local stationary-dynamics assumptions.

1.2. Contribution and organization.

Theorem 3.1 gives the two exact-dynamics storage criteria. Theorem 5.1 separates an arbitrary-resource channel statement from its thermal input; section 7 treats imperfect writing. The appendices give an infinite-bath interface, counterexamples and a separate collective-symmetry application. Data processing, Bell projection, channel twirling and noiseless subsystems are established machinery; no firstness is claimed for them. The contribution is the stated criteria and their proofs, with their dependencies explicit. A new local construction, efficient resource preparation and a noisy-writing threshold are separate tasks.

2. The mechanism in one page

Think of the memory as a logical qubit L, accessible auxiliary variables A, and a bath B that cannot be controlled. The auxiliary variables and bath may be correlated with each other and with the old logical qubit. An external input is swapped into L; the old logical qubit is discarded. The state left behind on $G = AB$ is its original marginal ω_G. This is a legitimate input-independent quantum channel.



Here $\mathfrak{a}$ is the sum of the three commutator norm bounds. The crucial operation is the fourth line. A general trace-distance estimate from relative entropy would introduce a square root of p. Instead, we ask only about the particular event P; its binary relative entropy controls a change from probability p to a constant multiple of p.

There is a second route. A Bell-state test turns a complete set of logical correlation estimates into entanglement fidelity. A random Pauli before writing, followed by its inverse after reading, converts that fidelity into an exact half-diamond error for the averaged channel. Thermal weighted correlations then give the bound $\mathfrak{a}p(2|t| + \beta)/4$. This protocol retains two classical frame bits reliably during storage. Neither route prepares ω or makes a nonlocal logical swap inexpensive.

3. Setup and the main storage theorem

3.1. **Access, resource and error criterion.** All Hilbert spaces in the main theorem are finite-dimensional. Let the accessible physical system have a specified subsystem identification

$$\mathcal{H}_S \simeq \mathcal{H}_L \otimes \mathcal{H}_A, \qquad \mathcal{H}_L = \mathbb{C}^2, \qquad \mathcal{H}_G = \mathcal{H}_A \otimes \mathcal{H}_B.$$

The identification must be implementable using the system alone. It need not be geometrically local or efficient. Let $H = H^*$ act on LG, and let Q_x, Q_y, Q_z be the ordinary Pauli matrices on L, extended by the identity. They satisfy the Pauli algebra on the *whole* physical Hilbert space, rather than only on a selected low-energy sector. We use units $\hbar = k_{\mathrm{B}} = 1$.

An external logical input, independent of a fixed resource state ω, is swapped into L. Tracing the old logical register gives

$$\mathcal{E}(\rho) = \rho \otimes \omega_G, \qquad \omega_G = \mathrm{Tr}_L\, \omega. \tag{3.1}$$

This identity holds also for inputs entangled with an arbitrary reference. No bath operation or conditional re-equilibration is involved. Holding and reading give the channel

$$\mathcal{N}_t(\rho) = \mathrm{Tr}_G[U_t(\rho \otimes \omega_G)U_t^*], \qquad U_t = e^{-itH}. \tag{3.2}$$

Reading means recovering the specified L factor and discarding A; the bath is inaccessible throughout.

For channels with the same input and output, write

$$d_\diamond(\mathcal{N}, \mathcal{M}) = \tfrac{1}{2}\|\mathcal{N} - \mathcal{M}\|_\diamond.$$

Equivalently, this is the supremum of half the output trace norm over reference-entangled inputs. This operational error lies in $[0, 1]$. In particular $\mathcal{N}_0 = \mathrm{id}_2$.

The randomized protocol chooses Q uniformly from $\{I, Q_x, Q_y, Q_z\}$, applies it to the external input before writing, and applies Q^* to the decoded output. Its channel is

$$\widetilde{\mathcal{N}}_t = \frac{1}{4}\sum_Q \mathrm{Ad}(Q^*) \circ \mathcal{N}_t \circ \mathrm{Ad}(Q), \qquad \mathrm{Ad}(V)(\rho) = V\rho V^*. \tag{3.3}$$

The resource and holding dynamics are independent of this choice. The guarantee for $\widetilde{\mathcal{N}}_t$ averages over the classical choice; it is not a guarantee for each fixed seed.

3.2. **An exact rare-boundary hypothesis.** Suppose

$$\omega = Z^{-1}e^{-\beta H}, \qquad Z = \mathrm{Tr}\, e^{-\beta H}, \quad 0 \le \beta < \infty. \tag{3.4}$$

Let P be an orthogonal projection and suppose, for $j = x, y, z$,

$$[P, Q_j] = 0, \qquad A_j := i[H, Q_j] = PA_jP, \qquad \|A_j\| \le a_j. \tag{3.5}$$

Define

$$p = \mathrm{Tr}(\omega P), \qquad \mathfrak{a} = a_x + a_y + a_z. \tag{3.6}$$

The two-sided support identity in (3.5) is stronger than a small Gibbs expectation of A_j^2. It says that the logical velocity vanishes exactly unless the state meets the boundary. We assume neither $[P, H] = 0$ nor $[P, \omega] = 0$.

For $K \ge 0$ set

$$F(K) = \inf_{\lambda > 0} \frac{K + e^\lambda - 1}{\lambda}. \tag{3.7}$$

Thus $F(0) = 1$, and the choice $\lambda = 1$ gives $F(K) \le K + e - 1$.

Theorem 3.1 (Storage from a thermal boundary)**.** *Under* (3.4)–(3.6)*, with the ideal access protocol* (3.1)–(3.2)*, for every real* t*,*

$$d_\diamond(\mathcal{N}_t, \mathrm{id}_2) \le \min\{1, \mathfrak{a} p |t| F(\beta\mathfrak{a}/4)\}, \tag{3.8}$$

$$d_\diamond(\widetilde{\mathcal{N}}_t, \mathrm{id}_2) \le \min\{1, \tfrac{1}{4}\mathfrak{a} p(2|t| + \beta)\}. \tag{3.9}$$

The randomized channel also obeys the bound in (3.8) *and*

$$d_\diamond(\widetilde{\mathcal{N}}_t, \mathrm{id}_2) \le \min\{1, \tfrac{1}{4} p t^2 (a_x^2 + a_y^2 + a_z^2)\}. \tag{3.10}$$

The minimum of these bounds may be used. In particular its error vanishes at $t = 0$*, although the right side of* (3.9) *alone has a nonzero constant term when* $\beta\mathfrak{a}p > 0$*.*

The proof of (3.8) is in section 4. The general channel theorem in section 5 and the thermal estimate in section 6 prove the remaining bounds. None of the constants contains the bath dimension. An estimate uniform in bath volume still requires p and a_j themselves to be uniform in that volume.

Corollary 3.2 (Conditional stretched-exponential storage)**.** *Suppose a sequence of models satisfies the theorem, and separately established estimates give*

$$p_n \le C n^r e^{-cn^\eta}, \qquad \mathfrak{a}_n \le C' n^s, \qquad 0 \le \beta_n \le \beta_* < \infty,$$

where $c, \eta, C, C' > 0$*,* $r, s \ge 0$ *and all constants are independent of* n*. For every* $0 < \varepsilon < c$*, both the deterministic and randomized errors tend uniformly to zero on*

$$|t| \le e^{(c-\varepsilon)n^\eta}.$$

Proof. The estimate $F(K) \le K + e - 1$ bounds the deterministic error by a polynomial in n times $e^{-\varepsilon n^\eta}$. The same is true of (3.9); twirling is also contractive in diamond norm. □

This is a consequence of a supplied equilibrium estimate, not a method for proving it. Geometry, disorder strength, interaction range, energy scales and a temperature interval enter through that separate task. In particular, bath-only terms vanish from the logical commutators but can still change p.

Example 3.3 (A two-qubit boundary that can be computed exactly)**.** Take one logical and one gauge qubit, $n = |1\rangle\langle 1|_G$, and

$$H = I_L \otimes \Delta n + g Q_z \otimes n.$$

Here $P = I_L \otimes n$, $a_x = a_y = 2|g|$, and $a_z = 0$. The actual Gibbs probability is

$$p = \frac{e^{-\beta\Delta}\cosh(\beta g)}{1 + e^{-\beta\Delta}\cosh(\beta g)}.$$

After replacement the gauge qubit has this same population, so

$$\mathcal{N}_t = (1-p)\,\mathrm{id}_2 + p\,\mathrm{Ad}(e^{-igtQ_z}).$$

Its half-diamond error is $p|\sin(gt)|$. Indeed, the two eigenvalues of e^{-igtQ_z} give the unitary-channel distance $|\sin(gt)|$, attained on an equatorial qubit; the mixture scales the channel difference by p. Pauli randomization removes the coherent Z rotation term and gives $\widetilde{\mathcal{N}}_t = (1 - p\sin^2(gt))\,\mathrm{id}_2 + p\sin^2(gt)\,\mathrm{Ad}(Q_z)$, with error $p\sin^2(gt)$. The exact error is at most p for all times. Thus the inverse-p storage interval in the theorem is a sufficient estimate, not an asserted optimal lifetime for every model.

4. Relative entropy keeps a rare boundary rare

The deterministic proof has two ingredients. First, a writer whose average output has relative-entropy cost $O(p)$ cannot increase the probability of a Gibbs event of weight p by more than a constant factor. Second, the support identities turn that probability directly into a bound on the logical channel derivative. We prove these statements before checking the thermal replacement writer.

For states σ, ω, with ω faithful, use natural logarithms and write

$$D(\sigma\|\omega) = \operatorname{Tr}\sigma(\log\sigma - \log\omega).$$

For $q, p \in (0,1)$, write

$$d(q\|p) = q\log(q/p) + (1-q)\log((1-q)/(1-p))$$

for binary relative entropy, with its continuous endpoint conventions. We use data processing under completely positive trace-preserving maps. Theorem 1 of Müller-Hermes and Reeb [11] implies the version needed here: relative entropy decreases under a positive trace-preserving map between finite-dimensional trace-class spaces. Our maps are the binary measurement and the partial trace of a finite classical flag, so they satisfy its hypotheses. The older completely positive result is due to Lindblad [8].

Lemma 4.1 (A binary rare-event bound)**.** *Let P be a projection, $p = \operatorname{Tr}(\omega P)$, and $q = \operatorname{Tr}(\sigma P)$. If $D(\sigma\|\omega) \le Kp$, then*

$$q \le pF(K). \tag{4.1}$$

Proof. Suppose first $0 < p < 1$. Measuring $\{P, I-P\}$ gives $d(q\|p) \le Kp$. For $\lambda > 0$, set

$$p_\lambda = \frac{pe^\lambda}{1-p+pe^\lambda}.$$

Expanding $d(q\|p_\lambda) \ge 0$ yields

$$\lambda q \le d(q\|p) + \log(1-p+pe^\lambda) \le p(K + e^\lambda - 1),$$

where $\log(1+x) \le x$ was used in the last step. Take the infimum over λ. For $p = 0$, faithfulness of ω implies $P = 0$; for $p = 1$, $q \le 1 \le F(K)$. □

The lemma asks only about one rare event. For example, $D(\sigma\|\omega) \le 10p$ gives $q \le (10 + e - 1)p$, however small p is. A general trace-distance estimate based on Pinsker's inequality would instead allow an additive error of order $\sqrt{p}$. That weaker estimate loses information relevant to the storage timescale.

Lemma 4.2 (Logical drift with an arbitrary reference)**.** *Suppose (3.5) holds and $\rho_{LGR}(t)$ follows the joint Hamiltonian evolution, with the identity on the reference R. Put $q_\rho(t) = \operatorname{Tr}(P\rho_{LG}(t))$ and $\rho_{LR}(t) = \operatorname{Tr}_G \rho_{LGR}(t)$. Then*

$$\tfrac{1}{2}\|\dot\rho_{LR}(t)\|_1 \le \tfrac{1}{2}\mathfrak{a}q_\rho(t). \tag{4.2}$$

Proof. Expand the logical qubit in its Pauli basis:

$$\rho_{LR}(t) = \tfrac{1}{2}I_L \otimes \rho_R + \tfrac{1}{2}\sum_{j=x,y,z} Q_j \otimes C_j(t).$$

The reference marginal is constant. Differentiating and using the Hamiltonian equation gives

$$\dot C_j(t) = \operatorname{Tr}_{LG}\left[(A_j \otimes I_R)\rho_{LGR}(t)\right].$$

This is Hermitian. For any Hermitian reference contraction B, the support and norm hypotheses imply the operator inequalities

$$-a_j P \otimes I_R \le A_j \otimes B \le a_j P \otimes I_R.$$

Consequently $|\operatorname{Tr} B\dot C_j(t)| \le a_j q_\rho(t)$. Trace-norm duality gives $\|\dot C_j(t)\|_1 \le a_j q_\rho(t)$. Since $\|Q_j\|_1 = 2$, the Pauli expansion proves (4.2). No bound depends on the reference dimension. □

Theorem 4.3 (An entropy condition for a writer)**.** *Let H act on a logical qubit and G, and let ω be a faithful state invariant under $U_t = e^{-itH}$. Assume (3.5). Let $\Gamma : \mathcal{B}(L)_* \to \mathcal{B}(LG)_*$ be a CPTP writing channel, and put $\sigma = \Gamma(I_L/2)$, $p = \operatorname{Tr}(\omega P)$. Suppose*

$$D(\sigma\|\omega) \le Kp, \qquad d_\diamond(\operatorname{Tr}_G \Gamma, \mathrm{id}_2) \le \varepsilon_0.$$

Then, for all real t,

$$d_\diamond(\operatorname{Tr}_G \operatorname{Ad}(U_t)\Gamma, \mathrm{id}_2) \le \min\{1, \varepsilon_0 + \mathfrak{a}p|t|F(K)\}. \tag{4.3}$$

Proof. Unitary invariance and stationarity give $D(U_t\sigma U_t^*\|\omega) = D(\sigma\|\omega)$. By lemma 4.1,

$$\operatorname{Tr}(PU_t\sigma U_t^*) \le pF(K) \quad \text{for every } t.$$

For any input marginal $\rho_L \le I_L$, positivity of Γ gives $\Gamma(\rho_L) \le \Gamma(I_L) = 2\sigma$. Thus every input, including one correlated with a reference, has physical boundary probability at most $2pF(K)$. Lemma 4.2 bounds the half trace norm of its decoded derivative by $\mathfrak{a}pF(K)$. Integrate from 0 to t, include the initial channel error ε_0, and take the reference supremum. □

This theorem concerns the actual joint output of the writer. It is a sufficient condition, not a necessary condition for useful storage. When p is exponentially small, the requirement $D(\sigma\|\omega) = O(p)$ is strong. It is not a constant-noise writing threshold.

4.1. **The Gibbs replacement calculation.**

Lemma 4.4 (Entropy cost of logical replacement)**.** *Under (3.4)–(3.6),*

$$\Omega := I_L/2 \otimes \omega_G = \tfrac{1}{4} \sum_{Q\in\{I,Q_x,Q_y,Q_z\}} Q\omega Q, \qquad D(\Omega\|\omega) \le \tfrac{1}{4}\beta\mathfrak{a}p.$$

Proof. The first equality is the Pauli depolarization identity on L; it can also be checked by expanding ω in the Pauli basis. The Gibbs formula and unitary invariance of entropy give

$$D(Q_j\omega Q_j\|\omega) = \beta \operatorname{Tr}\omega(Q_jHQ_j - H).$$

Now $Q_jHQ_j - H = -iQ_jA_j$. This operator is Hermitian, has norm at most a_j, and is supported on P on both sides because $[P, Q_j] = 0$. Therefore

$$D(Q_j\omega Q_j\|\omega) \le \beta a_j p. \tag{4.4}$$

To average these inequalities without an additional assumption, attach a classical flag to the four states $Q\omega Q$, with weights $1/4$, and compare with the flagged state having ω in every block. Their relative entropy is the average of the four relative entropies. Tracing the flag and using data processing proves the claimed inequality. □

Proof of (3.8) in theorem 3.1. Apply theorem 4.3 with $\Gamma = \mathcal{E}$, $\sigma = \Omega$, $K = \beta\mathfrak{a}/4$, and $\varepsilon_0 = 0$, using lemma 4.4. □

The same proof explains the role of the actual interacting Gibbs state. Replacing ω by an uncoupled reference state would generally destroy both stationarity and (4.4). Small interaction norm alone does not justify that substitution; see example B.6.

5. From correlations to a complete memory channel

The second route begins with a statement that needs no thermal assumption. The input is written into a logical factor of an arbitrary fixed resource. A complete operator basis tests how much that factor changes during a joint channel. A Bell projection converts these tests into a memory fidelity bound without assigning an input-dependent state to the bath.

5.1. **General logical dimension.** Let $L = \mathbb{C}^d$ and G be finite-dimensional, let ω be any state on LG, and let Λ be any CPTP map on LG. There is no rank, invariance or Gibbs assumption on ω. Define

$$\mathcal{N}(\rho) = \mathrm{Tr}_G\, \Lambda(\rho \otimes \omega_G).$$

Choose the discrete Weyl operators W_a, $a \in \mathbb{Z}_d^2$, with $W_0 = I$, $W_a^* W_a = I$, and $\mathrm{Tr}(W_a^* W_b) = d\delta_{ab}$. Explicitly they are the products of the cyclic shift and the diagonal phase with entries $1, e^{2\pi i/d}, \ldots, e^{2\pi i(d-1)/d}$. They form a projective group.

Let $\Phi = |\phi\rangle\langle\phi|$, where $|\phi\rangle = d^{-1/2} \sum_{j=1}^{d} |j\rangle_R |j\rangle_L$. The entanglement fidelity is

$$F_{\mathrm{e}}(\mathcal{N}) = \mathrm{Tr}\, \Phi\, (\mathrm{id}_R \otimes \mathcal{N})(\Phi).$$

Write Λ^* for the unital Heisenberg adjoint, and set

$$e_a = 2 - 2\,\mathrm{Re}\,\mathrm{Tr}\, \omega W_a^* \Lambda^*(W_a), \qquad \bar{e} = d^{-2} \sum_a e_a. \tag{5.1}$$

Theorem 5.1 (Bell–Weyl channel criterion)**.** *With the preceding definitions, each $e_a \geq 0$, and the complete-channel Weyl randomization*

$$\widetilde{\mathcal{N}} = d^{-2} \sum_a \mathrm{Ad}(W_a^*) \circ \mathcal{N} \circ \mathrm{Ad}(W_a)$$

satisfies

$$d_\diamond(\widetilde{\mathcal{N}}, \mathrm{id}_d) = 1 - F_{\mathrm{e}}(\mathcal{N}) \leq \min\{1, \bar{e}\}. \tag{5.2}$$

The resource and joint channel must be fixed independently of the randomization seed.

This is a complete input–output guarantee, including arbitrary references. Its bound has no dimension factor, even if d grows. The protocol still requires access to the logical algebra and a reliable classical frame with d^2 possible values. For $d = 2^k$, one may use the tensor-Pauli group and store exactly $2k$ frame bits.

5.2. **Why Bell compression resolves the resource correlation.** Before proving the theorem, consider what a Bell test does to the resource. Place an independent maximally mixed reference beside ω:

$$\Xi = I_R/d \otimes \omega, \qquad \Pi = \Phi_{RL} \otimes I_G.$$

Projecting RL onto its Bell state removes its old logical correlations and leaves the same marginal ω_G. The precise normalization is

$$d^2 \Pi \Xi \Pi = \Phi_{RL} \otimes \omega_G. \tag{5.3}$$

To verify it, write $\omega = \sum_{i,j} |i\rangle\langle j|_L \otimes \omega_{ij}$. The Bell matrix element of $I_R \otimes |i\rangle\langle j|_L$ is δ_{ij}/d; the additional factor $1/d$ in Ξ therefore leaves $d^{-2} \sum_i \omega_{ii}$. Equation (5.3) is a proof identity, not a proposed postselected writing operation. The physical writer remains the deterministic swap.

Proof of theorem 5.1. Let K_α be Kraus operators of Λ, extended by the identity on R, so $\sum_\alpha K_\alpha^* K_\alpha = I$. Using (5.3), the Bell-test failure is

$$\begin{aligned} 1 - F_{\mathrm{e}}(\mathcal{N}) &= d^2 \sum_\alpha \|(I - \Pi) K_\alpha \Pi \Xi^{1/2}\|_2^2 \\ &= d^2 \sum_\alpha \|(I - \Pi)[K_\alpha, \Pi] \Xi^{1/2}\|_2^2 \\ &\leq d^2 \sum_\alpha \|[K_\alpha, \Pi] \Xi^{1/2}\|_2^2. \end{aligned} \tag{5.4}$$

Here $\|\cdot\|_2$ is the Hilbert–Schmidt norm. The Bell expansion and normalized reference orthogonality are

$$\Pi = d^{-2} \sum_a \overline{W_a}_R \otimes W_a, \qquad d^{-1}\, \mathrm{Tr}(W_a^* W_b) = \delta_{ab}.$$

Substitution into the last line of (5.4) gives

$$d^{-2}\sum_{a,\alpha}\operatorname{Tr}\omega[K_\alpha,W_a]^*[K_\alpha,W_a].$$

For each a, expansion of the commutator yields

$$\sum_\alpha[K_\alpha,W_a]^*[K_\alpha,W_a]=2I-W_a^*\Lambda^*(W_a)-\Lambda^*(W_a^*)W_a.$$

Its expectation is e_a; this also proves $e_a\ge 0$. Thus $1-F_{\mathrm{e}}(\mathcal{N})\le\bar{e}$.

For completeness, Weyl randomization conjugates the normalized Choi matrix by $\overline{W_a}\otimes W_a$. The Weyl–Bell vectors $(I\otimes W_b)|\phi\rangle$ are joint eigenvectors with distinct characters. Averaging eliminates their off-diagonal entries. The resulting Choi matrix has diagonal probabilities $r_b\ge 0$, $\sum_b r_b=1$, and $r_0=F_{\mathrm{e}}(\mathcal{N})$. Its channel is $\sum_b r_b\operatorname{Ad}(W_b)$. The triangle inequality gives half-diamond error at most $\sum_{b\neq 0}r_b=1-r_0$. A Bell input gives orthogonal output components, and hence the matching trace-distance lower bound. This proves the equality in (5.2). □

An orthogonal unitary basis is enough for the Bell expansion, but not in general for the last randomization argument. The projective Weyl group, or the tensor-Pauli group, is part of the theorem's assumptions.

Corollary 5.2 (Exact stationary Hamiltonian dynamics)**.** *If* $\Lambda=\operatorname{Ad}(U_t)$, *put* $W_a(t)=U_t^*W_aU_t$. *Then*

$$e_a(t)=\operatorname{Tr}\omega[W_a(t)-W_a]^*[W_a(t)-W_a].$$

If additionally $[H,\omega]=0$, *define*

$$q=d^{-2}\sum_a\operatorname{Tr}\omega[H,W_a]^*[H,W_a].$$

Then $d_\diamond(\widetilde{\mathcal{N}_t},\mathrm{id}_d)\le\min\{1,t^2q\}$.

Proof. The first identity follows by expanding the square of the difference of two unitaries. For the second, use

$$W_a(t)-W_a=i\int_0^t U_s^*[H,W_a]U_s\,\mathrm{d}s.$$

Stationarity makes the weighted norm $\|X\|_\omega=(\operatorname{Tr}\omega X^*X)^{1/2}$ of every integrand equal to $\|[H,W_a]\|_\omega$. The triangle inequality for this seminorm proves $e_a(t)\le t^2\|[H,W_a]\|_\omega^2$. Apply theorem 5.1. □

Under the qubit hypotheses (3.5), $A_j^2\le a_j^2P$, so $q\le p\sum_j a_j^2/4$. This proves (3.10). Stationarity is needed for this initial-generator bound, although it is unnecessary in theorem 5.1 itself; example B.1 gives an explicit failure without it.

6. A linear-time thermal correlation estimate

The quadratic estimate just proved uses only the initial weighted commutator. Thermal structure yields a stronger long-time estimate. The useful correlation places $\sqrt{\omega}$ on each side of a logical operator. This symmetric placement lets the two support projections contribute $\sqrt{p}$ each, retaining p rather than $\sqrt{p}$.

Lemma 6.1 (A supported thermal Pauli correlation)**.** *Let* $Q=Q^*=Q^{-1}$, $\omega=Z^{-1}e^{-\beta H}$, *and*

$$[P,Q]=0,\qquad A=i[H,Q]=PAP,\qquad \|A\|\le a.$$

With $Q(t)=U_t^*QU_t$ *and* $p=\operatorname{Tr}\omega P$,

$$2-2\operatorname{Re}\operatorname{Tr}\omega QQ(t)\le ap(2|t|+\beta).\tag{6.1}$$

Proof. Define

$$C(t) = \operatorname{Tr} \omega Q Q(t), \qquad S(t) = \operatorname{Tr} \sqrt{\omega} Q \sqrt{\omega} Q(t), \qquad b = \|[Q, \sqrt{\omega}]\|_2^2.$$

The correlation $S(t)$ is real, since it is the trace of a product of Hermitian matrices. Expanding the commutator gives

$$b = 2 - 2S(0). \tag{6.2}$$

Stationarity moves the time evolution across the trace, so

$$S'(t) = \operatorname{Tr} \sqrt{\omega} Q(-t) \sqrt{\omega} A.$$

By the two-sided support of A, trace duality and the Hilbert–Schmidt product inequality give

$$\begin{aligned} |S'(t)| &\le a \|P \sqrt{\omega} Q(-t) \sqrt{\omega} P\|_1 \\ &\le a \|P\sqrt{\omega}\|_2 \, \|\sqrt{\omega} P\|_2 = ap. \end{aligned} \tag{6.3}$$

No commutation of P with ω was used.

We next compare S with the physical correlation C. Choose a common eigenbasis of H and ω, with Gibbs weights $w_m > 0$, and put $Q_{mn} = \langle m|Q|n\rangle$. Pairing m, n with n, m gives

$$\operatorname{Re} C(t) - S(t) = \tfrac{1}{2} \sum_{m,n} |Q_{mn}|^2 (\sqrt{w_m} - \sqrt{w_n})^2 \cos((E_m - E_n)t).$$

Consequently

$$|\operatorname{Re} C(t) - S(t)| \le b/2. \tag{6.4}$$

The same pairing in the relative entropy gives

$$D(Q\omega Q \| \omega) = \tfrac{1}{2} \sum_{m,n} |Q_{mn}|^2 (w_m - w_n)(\log w_m - \log w_n).$$

For $x, y > 0$,

$$(x - y)(\log x - \log y) \ge 4(\sqrt{x} - \sqrt{y})^2.$$

For example, writing $x/y = e^u$, $u \ge 0$, reduces this to $\tanh(u/4) \le u/4$; the case $u \le 0$ is symmetric. It follows that

$$2b \le D(Q\omega Q \| \omega) \le \beta a p, \tag{6.5}$$

where the last inequality is (4.4). Finally, (6.2) and (6.4) imply

$$2 - 2 \operatorname{Re} C(t) \le 2[S(0) - S(t)] + 2b \le 2ap|t| + \beta a p$$

by (6.3) and (6.5). □

Completion of theorem 3.1. Apply lemma 6.1 to the three Pauli matrices. The identity Weyl contributes zero, so theorem 5.1 gives

$$d_\diamond(\widetilde{\mathcal{N}}_t, \mathrm{id}_2) \le \tfrac{1}{4} \sum_{j=x,y,z} a_j p(2|t| + \beta),$$

which is (3.9). Convexity and unitary invariance of the diamond norm imply $d_\diamond(\widetilde{\mathcal{N}}_t, \mathrm{id}_2) \le d_\diamond(\mathcal{N}_t, \mathrm{id}_2)$; thus (3.8) also applies to the randomized channel. The quadratic bound was proved after corollary 5.2. □

The two proofs control different intermediate objects. The entropy proof controls the boundary occupation of the written state at every time. The thermal-correlation proof controls an equilibrium correlation and then uses a Bell test. Their hypotheses coincide in the main theorem, but the entropy-writer extension and the arbitrary-resource channel extension are separate statements.

7. Imperfect writing: sufficient conditions and obstructions

A memory that works after an ideal swap need not work after a noisy initialization circuit. The relevant error is a joint system–bath error, and an initially correct logical marginal is not enough. The entropy condition in theorem 4.3 is one sufficient interface. We now examine a different interface based on completely positive order, show its rigidity, and give a sector-conditioned repair.

7.1. Why domination by one ideal writer is restrictive.

For CP maps, write $\mathcal{F} \le_{\mathrm{CP}} \mathcal{E}$ when $\mathcal{E} - \mathcal{F}$ is CP. If an actual writer splits into a good subchannel dominated by a multiple of an ideal writer and a small bad subchannel, a positive failure test controls the output. However, the good component has an exact meaning here.

Proposition 7.1 (Rigidity of an ideal-writer subchannel)**.** *Let $\mathcal{E}_\gamma(\rho) = \rho \otimes \gamma$. If $\mathcal{F}$ is CP and $\mathcal{F} \le_{\mathrm{CP}} \kappa\mathcal{E}_\gamma$, then*

$$\mathcal{F}(\rho) = \rho \otimes \sigma, \qquad 0 \le \sigma \le \kappa\gamma.$$

In particular its success probability is independent of the logical input, and its normalized logical action is exact.

Proof. The normalized Choi operators obey $0 \le J_{\mathcal{F}} \le \kappa\Phi \otimes \gamma$. Their support on RL is therefore contained in the one-dimensional range of Φ. A positive operator with that support is $\Phi \otimes \sigma$, with $0 \le \sigma \le \kappa\gamma$. The Choi correspondence gives the asserted map. □

For example, a non-scalar logical rotation $\rho \mapsto U_\theta \rho U_\theta^* \otimes \gamma$ has no nonzero subchannel also dominated by $\mathcal{E}_\gamma$. The two rank-one logical Choi supports are distinct, so their intersection is zero. This remains true for arbitrarily small nonzero θ. Thus small diamond error does not imply a decomposition with a large CP-dominated good part.

Even faults that leave the logical factor exactly untouched can defeat domination by a single thermal gauge state.

Proposition 7.2 (Harmless faults with exponential domination cost)**.** *Let m gauge qubits have Hamiltonian $\Delta \sum_{i=1}^m n_i$, with $\Delta > 0, \beta > 0$, and no logical coupling. Put*

$$q = (1 + e^{\beta\Delta})^{-1}, \qquad \gamma_q = \mathrm{diag}(1-q, q).$$

After independent gauge bit flips of probability $0 < f < 1/2$, the writer is

$$\mathcal{W}_m(\rho) = \rho \otimes \gamma_r^{\otimes m}, \qquad r = q + f(1-2q) > q.$$

The logical channel is the identity for every holding time. Nevertheless, if

$$\mathcal{W}_m = \mathcal{F}_m + \mathcal{B}_m, \quad \mathcal{F}_m \le_{\mathrm{CP}} \kappa_m \mathcal{E}_{\gamma_q^{\otimes m}}, \quad \|\mathcal{B}_m^*(I)\| \le \delta < 1,$$

then for every $q < \theta < r$,

$$\kappa_m \ge \left(1 - \delta - e^{-md(\theta\|r)}\right) e^{md(\theta\|q)}. \tag{7.1}$$

Proof. Let B_m be the gauge event that at least $\lceil \theta m \rceil$ bits are one. The Chernoff bound for a Bernoulli variable, obtained by optimizing its exponential moment, gives

$$\mathrm{Tr}(\gamma_q^{\otimes m} B_m) \le e^{-md(\theta\|q)}, \qquad \mathrm{Tr}(\gamma_r^{\otimes m} B_m) \ge 1 - e^{-md(\theta\|r)}.$$

The integer rounding strengthens the respective upper and lower tail estimates. Evaluate the positive effect $I_L \otimes B_m$ on any input to the CP decomposition. Its good contribution is at most $\kappa_m e^{-md(\theta\|q)}$, and its bad contribution at most δ. Rearrangement proves (7.1). The logical identity follows directly from the absence of a logical coupling, before and after the gauge faults. □

This example uses bounded onsite energies and independent local faults. It disproves the necessity of global thermal domination as a general memory criterion. It does not disprove the usefulness of that criterion when a writer actually satisfies it.

7.2. **Condition on normalized sectors instead.** The preceding problem comes from a change in sector *weights*; the conditional state inside a sector can remain perfectly suitable. The following formulation separates these two issues.

Theorem 7.3 (Sector-conditioned admission)**.** *Let a CPTP writer have a finite or countable CP decomposition*

$$\mathcal{W} = \mathcal{B} + \sum_s \mathcal{F}_s, \qquad \mathcal{F}_s(\rho) = \rho \otimes \sigma_s, \qquad \operatorname{Tr} \sigma_s = w_s,$$

where $\|\mathcal{B}^*(I)\| \le \delta$*. For each* s*, suppose* γ_s *is a normalized state with*

$$\sigma_s \le \kappa_s w_s \gamma_s.$$

For a common holding-and-reading CPTP map $\mathcal{C} : LG \to L$*, assume*

$$1 - F_{\mathrm{e}}(\mathcal{C}\mathcal{E}_{\gamma_s}) \le \varepsilon_s.$$

Then the Weyl-randomized complete channel satisfies

$$d_\diamond(\widetilde{\mathcal{C}\mathcal{W}}, \mathrm{id}_d) \le \min\{1, \delta + \sum_s w_s \kappa_s \varepsilon_s\}. \tag{7.2}$$

Proof. On an encoded normalized Choi operator J, the failure functional is

$$\ell(J) = \operatorname{Tr}[(I - \Phi)(\mathrm{id}_R \otimes \mathcal{C})(J)].$$

It is positive and linear. The sector hypothesis gives $J_{\mathcal{F}_s} \le \kappa_s w_s \Phi \otimes \gamma_s$, hence $\ell(J_{\mathcal{F}_s}) \le \kappa_s w_s \varepsilon_s$. The bad Choi operator has trace at most δ, so its contribution is at most δ. Sum and use the Weyl-twirl identity $d_\diamond(\widetilde{\mathcal{C}\mathcal{W}}, \mathrm{id}_d) = 1 - F_{\mathrm{e}}(\mathcal{C}\mathcal{W})$, proved in theorem 5.1. For a countable decomposition, take monotone limits of the positive failure contributions. □

In proposition 7.2, condition on Hamming weight. Both $\gamma_q^{\otimes m}$ and $\gamma_r^{\otimes m}$, normalized inside a fixed-weight sector, are uniform on that sector. Thus one can take $\kappa_s = 1$, and every sector has $\varepsilon_s = 0$. The repaired criterion correctly gives zero error, regardless of the exponentially changed sector weights.

The repair still needs dynamics in the occupied sectors. In appendix C we give an example with a thermally rare sector that swaps the input into an inaccessible qubit. Independent gauge faults populate that sector with probability tending to one, although the logical input is exact immediately after writing. The example has a growing-support interaction and is therefore an obstruction to an abstract inference, not a local three-dimensional no-go theorem.

7.3. **What a local stochastic argument would have to prove.** A familiar cluster estimate can bound bad-sector probability once the necessary dynamical statement is available. We record its exact scope.

Proposition 7.4 (Union of thermal defects and writing faults)**.** *Let a graph have* n *vertices and maximum degree at most* $D \ge 2$*. Suppose random defect sets* T, F *satisfy, for every finite set* R*,*

$$\Pr(R \subset T) \le q^{|R|}, \qquad \Pr(R \subset F \mid T) \le f^{|R|} \quad \textit{almost surely.}$$

Then

$$\Pr(R \subset T \cup F) \le (q + f)^{|R|}.$$

The probability that $T \cup F$ *contains a connected component of size at least* ℓ *is at most*

$$n D^{2(\ell-1)} (q + f)^{\ell}. \tag{7.3}$$

Proof. Assign each site of R either to T or to F and use a union bound over assignments. Conditional probability bounds give a contribution at most $q^{|R_T|} f^{|R_F|}$; summing gives $(q + f)^{|R|}$. A connected component of size at least ℓ contains a connected ℓ-set. For a fixed root, a depth-first traversal of a spanning tree of such a set is a walk of length $2(\ell - 1)$; there are at most $D^{2(\ell-1)}$ such walks. Sum over roots and use the first bound. □

To turn (7.3) into a storage theorem one must additionally prove that the *whole configuration*, whenever all its components are smaller than ℓ, has a suitable conditional writer and safe dynamics. Correctability of each component in isolation is insufficient. Moreover, a noncommuting Gibbs state does not automatically define the classical defect law above, and a coherent good/bad amplitude decomposition is not automatically a CP instrument. No such local noisy-writing basin is established here.

8. Scope, comparisons and remaining questions

8.1. **What the theorem establishes.** Theorem 3.1 is a deterministic and randomized exact-dynamics storage criterion. It requires an accessible full logical algebra, exact two-sided commutator support and a rare-event estimate in the actual interacting Gibbs state. When that estimate is supplied, the logical error is linear in its probability. Corollary 3.2 retains the full exponential coefficient of that supplied probability, up to an arbitrarily small loss absorbing polynomial factors. It makes no optimality claim and does not produce a new equilibrium tail estimate.

The reusable part extends further than the thermal application. Theorem 5.1 works with an arbitrary fixed correlated state and an arbitrary joint CPTP channel. Appendix A proves its C^*-algebraic version, so an infinite bath need not have a global density matrix. This is an interface for a mathematical resource state and dynamics, rather than a finite-resource preparation of that state. A uniform finite-volume application must prove uniform estimates before passing to an infinite bath.

There is no implied stability under arbitrary perturbations. Changing the Hamiltonian can destroy (3.5), alter the equilibrium probability p, or make the chosen logical algebra inaccessible. A perturbation theorem must re-establish these hypotheses, or provide a different quantitative interface. A claim about a weak local coupling must specify whether its norm budget is per site, per logical commutator, or global, and which constants remain uniform.

8.2. **Other routes and their different resources.** Noiseless subsystems are a substantial alternative when symmetry is available. The representation-theoretic framework is classical [7, 6]; exact symmetry-protected Hamiltonian memories also occur in [16], Sections III.A–C. Appendix D explains a thermal use of four-spin singlets. Its bath may be shared, but the coupling is exactly collective within each cell. Generic single-site Pauli terms break the protection. Even the full-support logical algebra needs care: the highest-spin representation has multiplicity one, obstructing an exact unital logical qubit in the full collective commutant.

An alternative recent coding approach is the common-recovery construction in [12], Theorems 3.3 and 4.2 and Corollary 4.3. Its spatially varying noncommuting fields obey a single integrated operator-norm continuity modulus, uniform over the declared bath and field family. The physical sites densify in a fixed compact domain; a common operator envelope and decoupling argument give one encoder and a flag-dependent recovery family, with the erasure leading rate. This is an entanglement-transmission statement, with an independently supplied bath and an isometric code input. An array of independent local bath operators of fixed strength need not have the required continuity modulus. The result therefore supplies useful common-recovery machinery, but does not initialize the full-rank interacting Gibbs resource used in this paper or give a generic thermal noisy-writing theorem.

Active fault tolerance under exact non-Markovian noise is another essential comparator. Aharonov, Kitaev and Preskill [2] allow correlated bath dynamics under a sufficiently small summable interaction budget, with ongoing gates, resets and correction. Those controlled operations are resources. Conversely, an ideal-access passive storage estimate cannot claim a full-cycle advantage without counting its own preparation, writing, reading and classical control. All-time Markovian approximation results such as [10, Theorem 1.1] require specific reservoir regularity and spectral hypotheses; they do not replace the exact-dynamics argument for a general growing local spin bath.

8.3. **Decisive open operations.** The principal missing operation is a noisy writer into a dynamically admitted basin for an explicit local thermal model, with errors and costs uniform in device size. The sector formulation suggests a concrete target: identify observable sectors whose *normalized* conditional gauge–bath states remain controlled after a feasible writing process, and prove storage in all sectors receiving significant weight. The hot-sector example shows why the second part cannot be omitted.

A second target is replacing the exact support identities by a useful robust criterion. A uniform residual commutator bound ϵ typically accumulates as $\epsilon|t|$, so an exponentially long holding interval requires more than a fixed small perturbation estimate. One may instead seek a positive control observable $F \geq P$ with a small drift. Proposition B.7 records the resulting sufficient bound. Constructing such an F for a realistic imperfect writer remains open here.

Finally, model-dependent preparation and logical access must be quantified independently. A full logical algebra can be mathematically well defined yet require a costly global decoding operation. Neither a finite diagnostic nor the existence of a Gibbs state bounds those costs. A constructive resolution of these operations would turn the present storage criteria into a physical memory theorem with a complete resource account.

Acknowledgements and AI use

The mathematical ideas, constructions, proofs, and manuscript were developed by GPT-6 Astra, with additional AI assistance for review. Human input was minimal and consisted primarily of setting the research objectives and providing occasional feedback.

Appendix A. An algebraic interface for an infinite bath

The finite-dimensional logical channel remains meaningful even when a bath has no global density matrix. The right object is then a state on an observable algebra. We prove the extension directly, rather than take a formal trace over an infinite bath.

Theorem A.1 (Algebraic Bell–Weyl criterion)**.** *Let $\mathfrak{A}$ be a unital C^*-algebra with a specified accessible unital logical subalgebra M_d, let ω be any state on $\mathfrak{A}$, and let $T : \mathfrak{A} \to \mathfrak{A}$ be unital and completely positive. The logical matrix units identify $\mathfrak{A} \simeq M_d \otimes \mathfrak{B}$, where $\mathfrak{B}$ is the relative commutant. Let $\omega_{\mathfrak{B}}$ be the restriction of ω, and define the logical Heisenberg channel*

$$\mathcal{N}^*(a) = (\mathrm{id}_{M_d} \otimes \omega_{\mathfrak{B}})[T(a \otimes I)].$$

For standard logical Weyl operators, put

$$e_a = 2 - 2\,\mathrm{Re}\,\omega(W_a^* T(W_a)), \qquad \bar{e} = d^{-2} \sum_a e_a.$$

Then $e_a \geq 0$, and

$$d_\diamond(\widetilde{\mathcal{N}}, \mathrm{id}_d) = 1 - F_{\mathrm{e}}(\mathcal{N}) \leq \min\{1, \bar{e}\}.$$

No faithfulness, invariance or normality of ω is required.

Proof. First we describe the factorization and writer. If e_{ij} are the logical matrix units, the map

$$a \longmapsto [e_{1i} a e_{j1}]_{i,j}$$

identifies $\mathfrak{A}$ with matrices over the corner $e_{11}\mathfrak{A}e_{11}$; the inverse sends $[b_{ij}]$ to $\sum_{ij} e_{i1} b_{ij} e_{1j}$. The corner is isomorphic to the relative commutant by $b \mapsto \sum_i e_{i1} b e_{1i}$. This proves the stated tensor factorization. Swapping an external M_d with the logical matrix algebra uses the finite unitary $\sum_{ij} e_{ij}^{\mathrm{in}} \otimes e_{ji}$. After discarding the old register its state is $\rho \otimes \omega_{\mathfrak{B}}$, including for entangled inputs.

On $\mathfrak{C} = M_d^R \otimes \mathfrak{A}$, define the state $\Xi = \mathrm{tr}_R \otimes \omega$, with normalized trace, and the Bell projection $\Pi = \Phi_{RL} \otimes I$. Matrix-unit multiplication, as in (5.3), gives the state identity

$$d^2 \Xi(\Pi c \Pi) = (\Phi_{RL} \otimes \omega_{\mathfrak{B}})(c). \tag{A.1}$$

Let $(\pi, \mathcal{H}, \Omega)$ be the GNS representation of Ξ. We need a dilation of $\pi \circ (\mathrm{id}_R \otimes T)$. Here is the relevant Stinespring construction [15]. On the algebraic tensor product $\mathfrak{C} \odot \mathcal{H}$ use the positive semidefinite form

$$\langle c \otimes \xi, b \otimes \eta \rangle = \langle \xi, \pi[(\mathrm{id}_R \otimes T)(c^* b)]\eta \rangle.$$

Complete positivity proves positivity for finite sums. Quotient by the null space and complete. Left multiplication gives a representation π', bounded since $a^*a \leq \|a\|^2 I$ and the same matrix positivity applies. The map $V\xi = I \otimes \xi$ is an isometry by unitality, and

$$V^* \pi'(c) V = \pi[(\mathrm{id}_R \otimes T)(c)].$$

Put $p_0 = \pi(\Pi)$ and $p_1 = \pi'(\Pi)$. The encoded Bell state is represented, by (A.1), by the unit vector $dp_0\Omega$. Its output failure probability is

$$1 - F_{\mathrm{e}}(\mathcal{N}) = d^2 \|(I - p_1) V p_0 \Omega\|^2 \leq d^2 \|(V p_0 - p_1 V)\Omega\|^2.$$

Expanding the final square gives d^2 times

$$\Xi(\Pi) + \Xi[(\mathrm{id}_R \otimes T)(\Pi)] - 2\,\mathrm{Re}\,\Xi[\Pi(\mathrm{id}_R \otimes T)(\Pi)].$$

The first two terms equal d^{-2}: the normalized reference partial trace of Π is I/d^2, and T is unital. The Weyl expansion gives

$$\Xi[\Pi(\mathrm{id}_R \otimes T)(\Pi)] = d^{-4} \sum_a \omega(W_a^* T(W_a)).$$

Thus the square is exactly $\bar{e}$. For nonnegativity of each summand, Kadison's inequality yields

$$2I - W^* T(W) - T(W)^* W = [W - T(W)]^*[W - T(W)] + I - T(W)^* T(W) \geq 0.$$

Finally the logical channel acts on finite-dimensional M_d, so the Weyl-randomization argument in theorem 5.1 applies unchanged. □

Corollary A.2 (An invariant-state generator domain)**.** *Suppose α_t is a state-preserving automorphism group of $\mathfrak{A}$, implemented in the GNS representation of ω by a strongly continuous unitary group $U_t = e^{itK}$. Assume $\pi_\omega(W_a)\Omega_\omega \in \mathrm{Dom}(K)$ for every Weyl, and set*

$$q = d^{-2}\sum_a \|K\pi_\omega(W_a)\Omega_\omega\|^2.$$

For $T = \alpha_t$, the randomized logical channel has error at most $\min\{1, t^2 q\}$.

Proof. Invariance gives

$$e_a(t) = \|(U_t - I)\pi_\omega(W_a)\Omega_\omega\|^2 \le t^2\|K\pi_\omega(W_a)\Omega_\omega\|^2,$$

by $|e^{itx} - 1| \le |tx|$ and the spectral theorem. Apply theorem A.1. □

This domain condition is an $L^2(\omega)$ condition, not a claim that an infinite Hamiltonian exists as a bounded operator. The thermal inverse-rarity proof in theorem 3.1 was stated in finite dimension; the present appendix extends the general channel criterion and its quadratic invariant-state consequence, not every Gibbs entropy step.

Appendix B. Counterexamples to stronger inferences

The following small examples isolate hypotheses that can otherwise be hidden in informal memory arguments. All are finite-dimensional.

Example B.1 (Initial generator error without stationarity)**.** Let

$$H = I_L \otimes X_G + Q_z \otimes |1\rangle\langle 1|_G, \qquad \omega = I_L/2 \otimes |0\rangle\langle 0|_G.$$

Every logical Weyl satisfies $\mathrm{Tr}\,\omega[H, W]^*[H, W] = 0$, since its commutator is supported on $|1\rangle_G$. But ω is not invariant. In the two logical Z sectors the gauge Hamiltonians are

$$H_\pm = X \pm |1\rangle\langle 1| = \pm I/2 + X \mp Z/2.$$

At $t_* = 2\pi/\sqrt{5}$, their propagators are $-e^{\mp it_*/2} I_G$. Hence the logical channel is the nontrivial rotation $\mathrm{Ad}(e^{-it_* Q_z/2})$, with randomized error $\sin^2(\pi/\sqrt{5}) > 0$. The conclusion $t^2 q = 0$ would be false.

Example B.2 (Two Pauli correlations can miss complete loss)**.** Let $\omega = \Phi_{LG}$ and $H = \mathrm{SWAP}_{LG}$. The resource is invariant. Acting on $|\phi\rangle$, $X_L|\phi\rangle$ and $Z_L|\phi\rangle$ are symmetric under swap, while $Y_L|\phi\rangle$ is antisymmetric. Thus

$$e_x(t) = e_z(t) = 0, \qquad e_y(t) = 4\sin^2 t.$$

At $t = \pi/2$, replacement followed by the holding unitary swaps the input with the maximally mixed gauge qubit. The logical channel is completely depolarizing, with $F_{\mathrm{e}} = 1/4$ and half-diamond error $3/4$. Two correlations alone therefore cannot replace the complete-basis hypothesis of theorem 5.1 for a general correlated resource.

Example B.3 (Why randomization matters at growing dimension)**.** Let G be trivial, $\omega = I/d$, and $U = I - 2|0\rangle\langle 0|$, $d \ge 2$. A superposition of $|0\rangle$ and any orthogonal vector is mapped to an orthogonal state, so $d_\diamond(\mathrm{Ad}(U), \mathrm{id}_d) = 1$. On the other hand,

$$1 - F_{\mathrm{e}}(\mathrm{Ad}(U)) = 1 - \frac{|\,\mathrm{Tr}\,U|^2}{d^2} = \frac{4}{d} - \frac{4}{d^2}, \qquad \bar{e} = \frac{8}{d} - \frac{8}{d^2}.$$

The Weyl average in the last identity follows from $d^{-2}\sum_a W_a^* U W_a = (\mathrm{Tr}\,U/d)I$. Thus no dimension-independent untwirled interpretation of (5.2) is possible.

Example B.4 (Equal system and bath marginals are insufficient)**.** Take $H = gX_S \otimes Z_B$, $g > 0$, and $0 < c \leq 1$. The states

$$\rho_\pm = (I \otimes I \pm cY_S \otimes Z_B)/4$$

have the same maximally mixed marginals on both S and B. Since

$$e^{-itH}(Y \otimes Z)e^{itH} = \cos(2gt)Y \otimes Z + \sin(2gt)Z \otimes I,$$

their system outputs at $t = \pi/(4g)$ are $(I \pm cZ)/2$, whose trace-norm distance is $2c$. A marginal initialization error can therefore vanish while later system outputs differ by a fixed amount.

Proposition B.5 (An exactly consistent CP assignment is a product)**.** *If $\mathcal{A} : L \to LE$ is CPTP and $\mathrm{Tr}_E \mathcal{A} = \mathrm{id}_L$, then $\mathcal{A}(\rho) = \rho \otimes \tau_E$ for a fixed state τ_E.*

Proof. The normalized Choi state of $\mathcal{A}$ has pure marginal Φ_{RL}. A positive state with a pure marginal has support in that one-dimensional factor and therefore equals $\Phi_{RL} \otimes \tau_E$. Invert the Choi correspondence. □

This elementary fact is relevant to the correlated-initial-state discussion initiated by Pechukas [13]. It rules out assigning a different equilibrium bath state to every unknown input while preserving that input exactly by one CP map. The replacement writer (3.1) satisfies the proposition: its discarded factor is the fixed ω_G.

Example B.6 (Failure of an eventwise Gibbs comparison)**.** Let

$$H_0 = \Delta|1\rangle\langle 1|, \qquad V = \lambda X, \qquad P = |1\rangle\langle 1|, \qquad 0 < 2|\lambda| < \Delta.$$

As $\beta \to \infty$, the Gibbs probability of P for $H_0 + V$ converges to its ground-state value

$$\tfrac{1}{2}\left(1 - \frac{\Delta}{\sqrt{\Delta^2 + 4\lambda^2}}\right) > 0.$$

In contrast,

$$e^{2\beta\|V\|}\,\frac{\mathrm{Tr}(Pe^{-\beta H_0})}{\mathrm{Tr}\, e^{-\beta H_0}} = \frac{e^{2\beta|\lambda|}}{1 + e^{\beta\Delta}} \longrightarrow 0.$$

Thus a classical-looking multiplicative comparison of these event probabilities by $e^{2\beta\|V\|}$ is false for noncommuting Hamiltonians. Scalar spectral estimates for a whole invariant block remain valid and will be used in appendix D.

Proposition B.7 (A positive observable with small forward drift)**.** *Assume the logical support identities (3.5). Let $F \geq P$, $F \geq 0$, satisfy*

$$i[H, F] \leq \epsilon I, \qquad \sup_{\rho_L} \mathrm{Tr}[F\Gamma(\rho_L)] \leq \eta,$$

where $\epsilon, \eta \geq 0$ and Γ is a CPTP writer. If $d_\diamond(\mathrm{Tr}_G\, \Gamma, \mathrm{id}_2) \leq \varepsilon_0$, then, for $t \geq 0$,

$$d_\diamond(\mathrm{Tr}_G\, \mathrm{Ad}(U_t)\Gamma, \mathrm{id}_2) \leq \min\{1, \varepsilon_0 + \tfrac{1}{2}\mathfrak{a}(\eta t + \epsilon t^2/2)\}.$$

Proof. The Heisenberg derivative of the expectation of F is at most ϵ, so the boundary probability of any input at time $s \geq 0$ is at most $\eta + \epsilon s$. Integrate lemma 4.2 and include the release error. The supremum assumption concerns physical input marginals and hence also controls reference-entangled inputs. □

The one-sided drift inequality controls forward time. It is not a two-sided-in-time theorem unless a corresponding reverse inequality is available.

Appendix C. A rare sector made dangerous by independent writing faults

We give the full counterexample mentioned in section 7. It distinguishes an exact logical input at release from a dynamically admitted joint state. The deliberately nonlocal threshold interaction makes its scope transparent.

Fix $\Delta, \beta, g > 0$, let G contain m gauge bits, and let B be one inaccessible qubit. Write

$$q = (1 + e^{\beta\Delta})^{-1}, \qquad r = q + f(1 - 2q), \qquad 0 < f < 1/2,$$

and choose $q < \theta < r$. Let A_m project onto gauge strings of weight at least $\lceil \theta m \rceil$. On LGB set

$$H_m = \Delta \sum_{i=1}^{m} n_i + g A_m \otimes \mathrm{SWAP}_{LB}\,. \tag{C.1}$$

The norm of the second term is g, but its support grows with m. This is not a bounded-body local Hamiltonian family.

Proposition C.1 (Logical-neutral faults can destroy later storage)**.** *For the actual Gibbs state of (C.1), ideal logical replacement has half-diamond error at most $3\mu_m(A_m)/4$ at every time, where $\mu_m(A_m) \le Ce^{-md(\theta\|q)}$ and C is independent of m. After independent gauge bit flips of probability f at release, the logical input is still exact at time zero. At $t_* = \pi/(2g)$, however, every reader acting only on LG has half-diamond error at least $3/4 - o(1)$. The reader that simply returns L attains this limit.*

Proof. Let π_q be the product Bernoulli gauge law. For $a \in \{0, 1\}$, the trace of $e^{-\beta g a\,\mathrm{SWAP}}$ on LB is $3e^{-\beta g a} + e^{\beta g a}$, since the symmetric and antisymmetric multiplicities are three and one. Put

$$\alpha = (3e^{-\beta g} + e^{\beta g})/4.$$

The gauge law in the actual Gibbs state is

$$\mu_m(x) = \frac{\pi_q(x)\alpha^{A_m(x)}}{1 + (\alpha - 1)\pi_q(A_m)}. \tag{C.2}$$

Consequently

$$\mu_m(A_m) \le \frac{\alpha}{\min(1, \alpha)}\,\pi_q(A_m) \le Ce^{-md(\theta\|q)}.$$

For every gauge string, the reduced Gibbs state on B is $I_B/2$. Indeed the partial trace of both the identity and the swap is a scalar multiple of I_B. Ideal replacement therefore gives precisely

$$\rho_L \otimes \mu_m \otimes I_B/2.$$

In the cold sector the logical factor is unchanged. In the hot sector the unitary is $e^{-igt\,\mathrm{SWAP}}$. With a maximally mixed bath qubit it induces the logical channel

$$\rho \longmapsto \cos^2(gt)\rho + \sin^2(gt)I/2.$$

The half-diamond distance of a qubit depolarizing replacement $\rho \mapsto I/2$ from the identity is $3/4$: the Bell input attains it, and the Pauli-mixture representation gives the upper bound. The ideal overall error is hence $3\mu_m(A_m)\sin^2(gt)/4$, proving the first assertion.

Now apply independent gauge flips. Equation (C.2) shows that the total-variation distance of μ_m from π_q is

$$\frac{|\alpha - 1|\pi_q(A_m)(1 - \pi_q(A_m))}{1 + (\alpha - 1)\pi_q(A_m)} = O(\pi_q(A_m)).$$

Stochastic maps contract total variation. Thus the resulting gauge law μ'_m is $O(\pi_q(A_m))$-close to π_r. Since $r > \theta$, the lower-tail Chernoff bound gives

$$a_m := \mu'_m(A_m) \longrightarrow 1.$$

The post-fault state is still exactly $\rho_L \otimes \mu'_m \otimes I_B/2$.

At t_*, the hot-sector unitary is $-i\,\mathrm{SWAP}$. It sends the input into B, leaving all accessible variables independent of that input. On a Bell input, the reference marginal in this branch is $I_R/2$, and any accessible reader returns a state τ_L independent of R. Its Bell overlap is $\mathrm{Tr}[\Phi(I_R/2 \otimes \tau_L)] = 1/4$. The cold branch has Bell overlap at most one, so any reader has $F_{\mathrm{e}} \leq (1 - a_m) + a_m/4$. A Bell measurement then lower-bounds its half-diamond error by $1 - F_{\mathrm{e}} \geq 3a_m/4$. Returning L gives the mixture of identity and depolarizing replacement, with error exactly $3a_m/4$. □

The proposition does not show that every possible preparation fails. It shows that initial logical accuracy, independent gauge faults and an ideal equilibrium storage theorem do not by themselves imply storage after that noisy writer. In the sector criterion, the highly populated hot sector has large ε_s, so the failure is detected rather than hidden.

Appendix D. A separate application with exactly collective couplings

This appendix develops a symmetry-based application, independent of the rare-boundary theorem. It illustrates both the usefulness and the limits of importing representation theory into a thermal memory problem. Noiseless subsystems and collective-noise singlets are established constructions [7, 6]. The thermal estimates below make their resource assumptions explicit.

D.1. **A full-support obstruction.** For n spin-1/2 particles, the collective $SU(2)$ representation decomposes as

$$(\mathbb{C}^2)^{\otimes n} = \bigoplus_j V_j \otimes M_j,$$

with irreducible spin spaces V_j and multiplicity spaces M_j. The algebra commuting with all collective rotations is $\bigoplus_j I_{V_j} \otimes \mathcal{B}(M_j)$. The maximal-spin summand has $\dim M_{n/2} = 1$. Indeed, averaging an inner product over the compact group gives a unitary representation and complete reducibility; Schur's lemma then gives the displayed commutant. The weight-$n/2$ space is spanned by the all-up vector, so only one maximal-spin copy can occur.

Proposition D.1 (No exact full-support qubit in the collective commutant)**.** *There is no unital embedding of M_2 in this commutant. Consequently, for a faithful state, an exact logical qubit on the full Hilbert space cannot have zero weighted commutator with every collective spin operator.*

Proof. Projection onto the maximal-spin multiplicity gives a unital character from the commutant to $\mathbb{C}$. A unital embedding of M_2 would compose with it to give a unital homomorphism $M_2 \to \mathbb{C}$, which is impossible: its kernel must be either zero or all of the simple algebra M_2, and an injective map into the commutative algebra $\mathbb{C}$ cannot exist. For a faithful state, $\omega(X^*X) = 0$ implies $X = 0$. Thus zero weighted commutators imply exact operator commutation and the same obstruction. □

Restriction to a singlet sector avoids the multiplicity-one summand, but a finite-temperature faithful Gibbs state has support outside that sector. One must account for those flags.

D.2. **A conditional Gibbs estimate for singlet cells.** Partition $4N$ spins into N four-spin cells. The total-spin-zero subspace of a cell has dimension two; let P_b project onto it and write $Q_b = I - P_b$, of rank fourteen. The dimension follows by coupling two pairs: $(V_0 \oplus V_1) \otimes (V_0 \oplus V_1)$ contains one singlet from $V_0 \otimes V_0$ and one from $V_1 \otimes V_1$. Consider a finite bath and a Hamiltonian

$$H = H_B + \Delta \sum_{b=1}^{N} Q_b + \sum_{b=1}^{N} F_b + H_{\mathrm{int}}. \tag{D.1}$$

Here H_B acts on the bath alone. Each F_b is supported on cell b and the bath, is collective within that cell, satisfies $F_b P_b = 0$, and has norm at most M. Different F_b may use the same bath. Write $H_{\mathrm{int}} = \sum_X h_X$, and assume

$$\sum_{X:\, b\in X} \|h_X\| \le J \quad \text{for every cell } b.$$

Every term in (D.1) preserves every flag P_b, Q_b. The h_X may include bath variables, but their indicated norm budget includes all terms incident to b.

Proposition D.2 (Thermal domination of bad flags)**.** *For $\beta > 0$, the joint Gibbs law of the flags in* (D.1) *is stochastically dominated by independent Bernoulli bad flags of parameter*

$$r = 7e^{-\beta(\Delta - M - 2J)}, \qquad u = \frac{r}{1+r}. \tag{D.2}$$

In particular $\Pr(Q_b = 1 \mid \textit{all other flags}) \le u$.

Proof. Fix the flags at every cell other than b. Remove F_b, the penalty at b, and every interaction term incident to b. The remaining Hamiltonian K is independent of the spin variables at b, even when bath variables are shared. Let Z_K be its partition function in the fixed other-flag sector. The removed incident interaction has norm at most J. On the good block the partition function is at least

$$Z_{\mathrm{good}} \ge 2e^{-\beta J} Z_K.$$

On the bad block F_b contributes norm at most M, so

$$Z_{\mathrm{bad}} \le 14 e^{-\beta\Delta + \beta(M+J)} Z_K.$$

These scalar trace bounds follow from the min–max eigenvalue inequalities for a Hermitian perturbation of bounded norm. They require no eventwise operator comparison of noncommuting Gibbs states. The conditional bad-to-good odds are at most their ratio, namely r, and the probability is at most u.

Conditioning on only an initial subset of flags gives the same upper bound by conditional expectation. Sample flags successively using independent uniforms U_b; declare the actual flag bad if U_b is below its conditional probability. It is then contained in the independent bad flag $\mathbf{1}_{\{U_b \le u\}}$. This proves stochastic domination without an FKG assumption. □

D.3. **Ideal routing and a supplied outer memory.** When $H_{\mathrm{int}} = 0$, every good cell carries a two-dimensional singlet subsystem on which all F_b vanish. One can store in these cells forever under the stipulated collective dynamics. The flags need not be independent; proposition D.2 suffices.

Corollary D.3 (Conditional positive-rate storage with ideal access)**.** *Assume $H_{\mathrm{int}} = 0$. For $k \le N$, there is a specified full-support M_{2^k} factor on the physical spins and an ideal system-only replacement and reading protocol whose half-diamond error, at every time, is at most*

$$\Pr(\textit{fewer than } k \textit{ good cells}).$$

If $k \le (1 - u - \zeta)N$, with $0 < \zeta < 1 - u$, this probability decays exponentially in N. No efficient or local implementation of this routing is asserted.

Proof. In a flag sector with $g \ge k$ good cells, choose the first k singlet factors as the logical system. The Hamiltonian is the identity on those factors. A sector with g good cells has spin dimension

$$2^g 14^{N-g} = 2^N 7^{N-g},$$

which is divisible by 2^k even if $g < k$. Choose any 2^k-dimensional factor in these unsuccessful sectors. Taking the direct sum of these sector identifications gives one full-space factorization with the flags in its gauge factor. The logical swap preserves their Gibbs probabilities. On each successful sector the logical channel is exactly the identity, even if the other variables

and bath are correlated. On an unsuccessful sector it is some CPTP channel. Convexity bounds half-diamond error by the unsuccessful probability. Finally proposition D.2 bounds the number of bad cells by a $\mathrm{Bin}(N, u)$ variable. Its upper-tail exponential-moment bound proves the last statement. □

This corollary does not contradict proposition D.1. The full-space logical operators commute with the dynamics in successful sectors; their arbitrary extension in unsuccessful sectors need not commute with all collective fields.

One may instead supply an outer Hamiltonian and memory protocol on one qubit per cell. Let $V_b : \mathbb{C}^2 \to \operatorname{ran} P_b$ be singlet isometries. Lift each outer term h_X by $\bigotimes_{b\in X} V_b$ and extend it by zero whenever a cell in X is bad. This preserves its norm, number of incident cells and body size, up to the fixed four-spin cell realization. In the all-good sector, the actual Hamiltonian is the outer Hamiltonian plus H_B; all F_b vanish. Its normalized Gibbs resource is exactly the outer resource times the bath Gibbs state.

Corollary D.4 (Conditional outer-memory lift)**.** *For the lifted terms just described, suppose the supplied outer memory has error at most $\varepsilon_{\mathrm{out}}(t)$ with its stated ideal resource and access. A flag-preserving protocol that uses it in the all-good sector has error at most*

$$\varepsilon_{\mathrm{out}}(t) + Nu.$$

For $0 < \delta < 1$, the choice

$$\Delta \geq M + 2J + \beta^{-1} \log(7N/\delta)$$

makes the additional error at most δ.

Proof. The probability of any bad flag is at most Nu by proposition D.2 and a union bound. The all-good conditional state and dynamics are the supplied outer ones; the remaining branches contribute at most their probability in half-diamond distance. The displayed penalty gives $r \leq \delta/N$ and hence $Nu \leq \delta$. □

The penalty grows as $\log N$ at fixed temperature and error budget. It is a change in physical resources, not a fixed-energy-scale thermodynamic stability theorem. For a different retained cell dimension d_b inside n_b spins, the rank factor 7 in (D.2) becomes $(2^{n_b} - d_b)/d_b$, or its maximum over cell types. For example, a seven-dimensional subspace of an eight-spin singlet sector gives 249/7.

D.4. **Why a large penalty alone does not protect against a bath.** Within a singlet sector, $P_b \sigma_i^z P_b = 0$, while $\|\sigma_i^z |s\rangle\| = 1$ for a unit singlet $|s\rangle$. The first identity follows because a vector operator has zero matrix elements between total-spin-zero states; equivalently, averaging the rotated σ_i^z over $SU(2)$ gives zero. Put $|v\rangle = \sigma_i^z |s\rangle$, which lies in the bad sector. Take a bath qubit with Hamiltonian $\Delta |1\rangle\langle 1|_B$ and add the symmetry-breaking coupling

$$\lambda \sigma_i^z \otimes X_B.$$

The subspace spanned by $|s\rangle|1\rangle_B$ and $|v\rangle|0\rangle_B$ is invariant. Both unperturbed energies equal Δ, and the coupling is λX on this subspace. Leakage is therefore $\sin^2(\lambda t)$, reaching one independently of Δ. This does not describe the probability of this bath input in a specified thermal state; it disproves a state-independent claim that the gap alone suppresses leakage against an arbitrary resonant bath.

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